

ai-os-new

Observability Re-Audit Report

Verification audit after bug fixes. Original: 21 findings. Remaining: 21 findings.

Severity	Count
CRITICAL	3
HIGH	8
MEDIUM	7
LOW	3
Total	21

OBS
-01

CRITICAL

ErrorBoundary emits to EventBus but never to ILogger

Files: ErrorBoundary.tsx

Problem: componentDidCatch only fires EventBus notification (transient toast). Never writes to LoggerService. Stack trace and component tree info discarded.

Fix: Import rootLogger and call rootLogger.error(). Also emit dedicated ERROR_BOUNDARY_CAUGHT event.

OBS
-02

CRITICAL

ProviderInstance errorCount/successCount cumulative - health permanently stuck

Files: provider-instance.ts

Problem: getHealth() compares errorCount > successCount*2. After recovery, ratio still >2x due to historical errors. Key permanently flagged unhealthy.

Fix: Replace cumulative counters with sliding-window ring buffer or EWMA. Add resetCounters() on recovery.

OBS
-03

CRITICAL

MetricsService successRate is provider-status ratio, not request success

Files: metrics-service.ts

Problem: successRate = activeProviders/totalProviders. Not actual request success rate. Dashboard shows 100% success even when all requests fail.

Fix: Track actual request-level success/failure. Replace successRate/errorRate with real request metrics.

OBS
-04

HIGH

FeatureFlagService mutations have zero audit trail

Files: feature-flag-service.ts

Problem: setEnabled() calls local listeners but never emits to event bus, logs change, or records who/when.

Fix: Add EventBus.emit(EVENTS.FEATURE_FLAG_CHANGED). Add ILogger dependency.

OBS
-05

HIGH

ConfigService mutations have zero event emission

Files: config-service.ts

Problem: All updateX() methods mutate CONFIG and persist overlays but never emit events. ConfigHistoryService never called from ConfigService.

Fix: After setConfig+persist, emit EVENTS.CONFIG_CHANGED. Integrate ConfigHistoryService.commit().

OBS
-06

HIGH

KeyLifecycle transitions never emitted to event bus

Files: key-lifecycle.ts

Problem: transition() only pushes to internal transitions[] array. No event bus emission. No consumer can observe lifecycle state changes in real time.

Fix: Add EventBus.emit(EVENTS.KEY_LIFECYCLE_TRANSITION, {keyId, from, to, reason}).

OBS
-07

HIGH

HealthService.writeToKeyStateStore has silent catch

Files: health-service.ts

Problem: catch { /* silent */ } - DB write failures invisible. HealthService thinks key is healthy but state store is stale.

Fix: Replace with catch(e) { logger.warn(...); }. Add ILogger.

OBS
-08

HIGH

Cross-tab sync has no desync detection

Files: cross-tab-state.ts

Problem: handleSyncResponse blindly overwrites local state. No comparison with local. If divergent, no detection or logging.

Fix: Compare received vs local state. Emit CROSS_TAB_DESYNC event if divergent.

OBS
-09

HIGH

CacheService hit/miss counters cumulative - hitRate meaningless

Files: cache-service.ts

Problem: hits/misses monotonically increasing. After days, sudden drops in effectiveness take very long to appear.

Fix: Add sliding-window counter or periodic reset. Add lastResetAt to getStats().

OBS - 10	HIGH	ProviderTracker.hydrateState() silent catch
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Files: provider-tracker.ts

Problem: catch { /* silent */ } - DB read failure means tracker starts with empty state. All accumulated data lost with zero notification.

Fix: Replace with catch(e) { console.error(...); }. Emit monitoring event.

OBS - 11	HIGH	ConfigService.init() silent catch
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Files: config-service.ts

Problem: catch { this.overlays = {}; } - DB load failure silently falls back to defaults. Users' custom configs vanish.

Fix: Replace with catch(e) { console.error(...); this.overlays = {}; }.

OBS - 12	MEDIUM	ExecutionQueue has zero observability
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Files: execution-queue.ts

Problem: Tracks pending/inflight but never emits events for enqueue/start/complete/fail. No way to monitor queue depth or latency.

Fix: Add event emissions for queue lifecycle. Add ILogger for high depth warnings.

OBS - 13	MEDIUM	DebateRoom state changes have no event bus emission
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Files: debate-room.ts

Problem: applyOverride/injectEvent use private callbacks but never emit to event bus. Lifecycle changes (start/pause/resume/stop) also silent.

Fix: Add event bus dependency. Emit events for all state mutations.

OBS - 14	MEDIUM	ProviderRuntimeState is poll-only - no event-driven updates
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Files: provider-state.ts

Problem: Only notifies via polling (10s). Critical state changes not immediately emitted. Dead instance not visible for up to 10s.

Fix: Emit events on significant transitions. Add EventBus to constructor deps.

OBS - 15	MEDIUM	ProviderBudget exhaustion invisible to monitoring
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Files: provider-budget.ts

Problem: canStartSession/canProceed returning {allowed:false} is never emitted as event. exhausted flag computed on-demand but no event fires on transition.

Fix: Add EventBus. Emit BUDGET_EXHAUSTED when exhausted transitions true.

OBS - 16	MEDIUM	Zustand stores have no state-change telemetry
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Files: debateLiveStore.ts, topologyTraceStore.ts

Problem: No middleware to log state changes or detect anomalies. If bug causes 10k events in 1s, no early-warning signal.

Fix: Add Zustand middleware that logs count changes at WARN level when growth exceeds threshold.

OBS - 17	MEDIUM	VirtualKeyService multiple silent catches
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Files: virtual-key-service.ts

Problem: Three catch blocks (init, getRealKey, doPersist) silently swallow errors. Persistence failures invisible.

Fix: Add ILogger. Log warnings on persistence failures.

OBS - 18	MEDIUM	WorkspaceService multiple silent catches
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Files: workspace-service.ts

Problem: Multiple catch blocks (persistHandle, tryRestoreHandle, traverseDir, grepDir) silently swallow errors.

Fix: Add ILogger. Log warnings for all error cases.

OBS -19	LOW	UsageTracker uses console.warn instead of ILogger
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Files: `usage-tracker.ts`

Problem: All error logging uses `console.warn`. Won't appear in `LogsPanel`, can't be filtered.

Fix: Add `ILogger` dependency. Replace `console.warn` with `logger.warn()`.

OBS -20	LOW	RaceExecutor has zero observability
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Files: `race-executor.ts`

Problem: Tracks failures in return value but never emits events or logs. All-candidates-fail scenario produces no structured data.

Fix: Add event bus emission for `RACE_COMPLETED/RACE_ALL_FAILED`. Add `ILogger`.

OBS -21	LOW	CompromiseWebhookService has no ILogger
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Files: `compromise-webhook-service.ts`

Problem: Handles security-critical compromise signals but no structured logger. Malformed payloads return false silently.

Fix: Add `ILogger`. Log received payloads, parsing failures, compromise signals.